JX JX Runtime + Onyx Engine

Onyx runs the model. JX guards the door.



TWO PROJECTS

Execution and access, kept apart.

JX Runtime

The local model runtime you can expose to someone other than yourself.

- Access keys & pairing
- TLS & network posture
- Model installs & fit checks
- Crash detection & restart
- Metrics without content
- Audit log

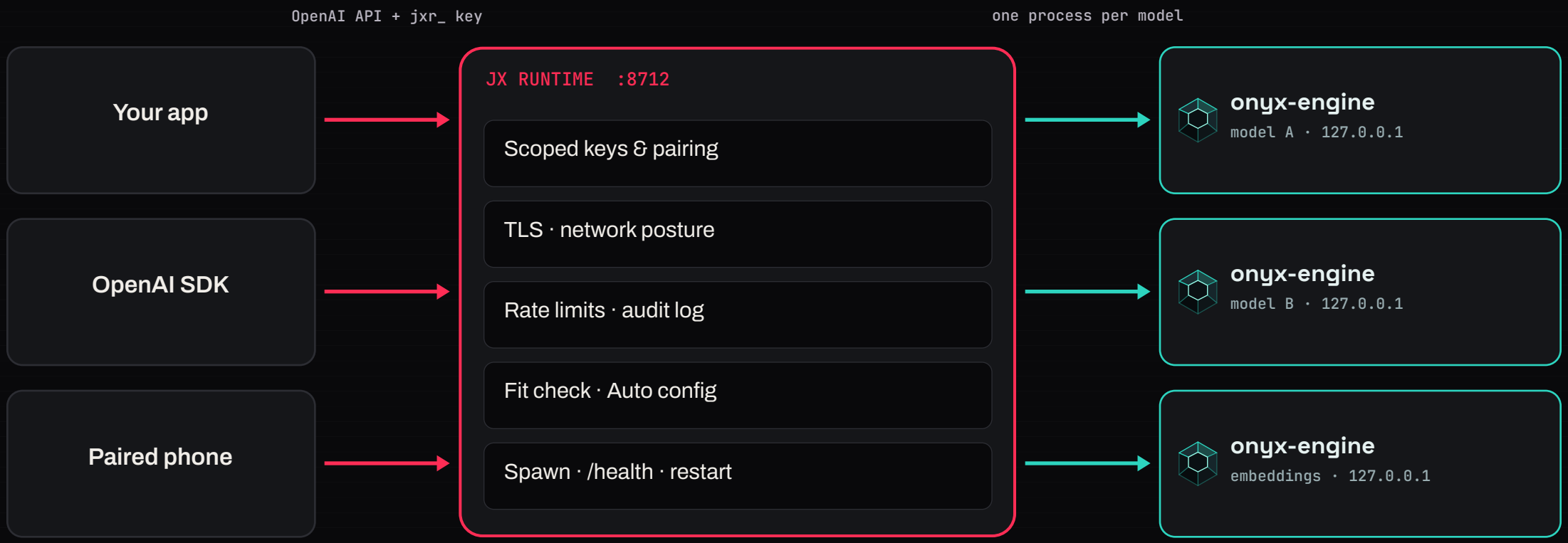
Onyx Engine

A single-model, OpenAI-compatible server built on llama.cpp.

- Inference on llama.cpp
- Continuous batching
- Streaming over SSE
- Tool calls & grammars
- Embeddings
- Multimodal input

HOW THEY FIT

One endpoint. One engine per model.



SWITCH ENGINES

Four steps. One setting.

01 Install

```
jx-runtime backend
install --engine onyx
```

A prebuilt onyx-engine, checked against its release checksums, lands in the data directory.

02 Select

```
execution.backend:
"onyxengine"
```

One setting. Its adapter subclasses the llama.cpp one: same spawn, streaming and restart code.

03 Serve

```
one onyx-engine per
loaded model
```

Bound to 127.0.0.1, polled at /health, its OpenAI endpoints proxied through.

04 Share

```
Authorization: Bearer
jxr_…
```

Apps and paired devices reach it through JX Runtime's keys, TLS and rate limits.

SIDE BY SIDE

Compare engines on the same models.

terminal

```
$ jx-runtime dual-engine plan      # prints, changes nothing
$ jx-runtime dual-engine setup --yes

$ jx-runtime serve                                  # :8712  llama.cpp
$ jx-runtime serve --data-dir ~/.jx-runtime-onyx    # :8713  Onyx Engine
```

The secondary borrows the primary's model folders. Keys, pairing and the audit log are per instance and never copied.

NO SECOND COPY

Same contract. Same code path.

`/health`	Polled until ready. For onyx-engine, ready means the model is loaded.
`/props`	Context, template hash and build info, read once after start.
`/v1/*`	Chat, completions and embeddings proxied straight through, streaming included.
`restart`	Crash detection and failure classification are the llama.cpp adapter's own code.

Run it on your machine.
Share it on your terms.

JX Runtime

`github.com/jxburros/JX-Runtime`

Alpha. Release binaries are not yet code-signed.

Onyx Engine

`github.com/JX-Holdings-LLC/Onyx-Engine`

v0.4.0. Prebuilt for linux-x64, darwin-arm64, darwin-x64 and win32-x64.